

Pose-Based Isolated Sign Recognition in the Browser: Signer-Independent Evaluation and Cross-Lingual Transfer from ASL to French Belgian Sign Language

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Abstract

Pose-based isolated sign recognition is commonly evaluated on random splits and rarely deployed on consumer hardware. We study it under the condition that matters in practice: recognising signs produced by people absent from the training data. A single preprocessing function, shared by training and by a JavaScript port, converts MediaPipe Holistic landmarks into features for a 2.2 M-parameter convolution–Transformer. On Google ISLR (250 American Sign Language signs, 21 signers), the model reaches $64.8 \pm 0.4\%$ top-1 accuracy on four unseen signers, against $63.0 \pm 0.5\%$ for a bidirectional GRU and 42.1% for a statistical baseline; a random split overestimates this accuracy by 16.9 points. One-factor ablations identify the canonical dominant hand as the most influential preprocessing step. On LSFB-ISOL (100 glosses of French Belgian Sign Language), initialising the model from ASL brings no gain at any training-set size, from 5 examples per sign to the full corpus and is slightly but significantly harmful with all data (66.2% against 68.3% from scratch on 40 unseen signers). We also document three silent failure modes of such pipelines: augmentations cancelled by normalisation, anisotropic coordinates in anamorphic video, and feature outliers that break INT8 quantisation. In the browser, the model classifies a sign in 16 ms on one CPU thread and reproduces the PyTorch predictions on every tested clip.

1 Introduction

Sign languages are natural languages with their own lexicon and grammar, used by millions of Deaf and hard-of-hearing people. Automatic sign language processing covers tasks of increasing difficulty: *isolated* sign recognition (one sign per clip, closed vocabulary), continuous recognition, and translation to and from spoken languages [5, 18, 31]. This report addresses isolated recognition only. The distinction is not merely terminological: a classifier over a few hundred glosses ignores the grammar of the language, its use of space and its non-manual markers, and therefore cannot translate [31].

Pose-based approaches first extract body, hand and face landmarks with an off-the-shelf estimator such as MediaPipe Holistic [2, 17, 21, 33], then classify the resulting keypoint sequence. Compared with models operating on RGB video, they are compact, largely insensitive to background and appearance, and allow inference without transmitting video, which protects the privacy of users. The Google Isolated Sign Language Recognition (ISLR) competition [8] established this setting with 94,477 landmark sequences of 250 American Sign Language (ASL) signs.

Three questions remain insufficiently addressed in this setting. They structure the report.

RQ1 Generalisation. How accurately do pose-based models recognise signs produced by people absent from the training data, and how much does a random split overestimate this accuracy?

RQ2 Transfer. Does pre-training on ASL, a comparatively well-resourced sign language, improve recognition in French Belgian Sign Language (LSFB), in particular when few labelled examples are available?

RQ3 Deployment. Can the complete pipeline run in real time in a web browser on a laptop CPU while reproducing the predictions of the trained model?

Contributions.

- A single, unit-tested preprocessing function shared by training, evaluation and a JavaScript port, whose outputs match the reference implementation within 3×10^{-6} (Section 4).
- A signer-independent comparison of a statistical baseline, a bidirectional GRU and a convolution–Transformer on 250 ASL signs over three seeds, with paired significance tests, a measurement of split leakage and one-factor ablations (Sections 6.1 and 6.2).
- A controlled transfer study from ASL to LSFB, with learning curves from 5 examples per sign to the full corpus and a linear probe of the pre-trained encoder (Section 6.3).
- A fully client-side demo that classifies a sign in 16 ms, and an analysis of INT8 quantisation, which first degrades accuracy because of feature outliers and, once corrected, remains slower on CPU (Section 6.4).

Section 2 reviews related work. Sections 3 to 5 describe the data, the method and the evaluation protocol. Section 6 reports the results, Section 7 discusses them together with their limitations, and Section 8 answers the three research questions.

2 Related work

Isolated sign recognition datasets. Large ASL vocabularies are available in WLASL [19] (2,000 glosses collected from the web), MS-ASL [28] and ASL Citizen [10]; the latter is community-sourced, collected with explicit consent and provided with signer-independent splits. BSL-1K [1] mines British Sign Language signs from broadcast footage using mouthing cues. The Google ISLR competition [8] released MediaPipe landmarks only, recorded by 21 participants on smartphones, which removes the need to store video. For sign languages related to French, LSFB-ISOL and LSFB-CONT [12] are derived from the LSFB corpus [23]. On LSFB-ISOL, Fink et al. [13] train a two-layer Transformer on MediaPipe landmarks of 700 glosses with more than 20 examples each, under a signer-independent split, and report 54 % top-1 and 83 % top-10 accuracy. No subset of LSFB-ISOL has become a standard benchmark, so our results on the 100 most frequent glosses cannot be compared directly with theirs. MEDI-API-SKEL [6] and Dicta-Sign-LSF-v2 [3] provide continuous French Sign Language (LSF) with skeletons or annotations but no isolated-sign benchmark.

Pose-based models. Skeleton sequences have been classified with spatio-temporal graph convolutions [16, 27, 30], with Transformers operating directly on keypoints [4], and with self-supervised pre-training of hand representations [15]. The strongest solutions of the Google ISLR competition combined depthwise temporal convolutions with a few self-attention layers, a design close to the Conformer used in speech recognition [14], trained for several hundred epochs with strong regularisation and averaged over ensembles of models. Our convolution–Transformer follows this design at a smaller scale. Recurrent networks [7] remain a light and competitive baseline.

Transfer across sign languages. Sign languages share articulatory parameters (handshape, movement, location), which suggests that features learned on a large corpus may transfer to a smaller one [32]. Large-scale pre-training improves recognition on smaller datasets of the *same* language [1]. Transfer between distinct sign languages from pose data alone has received less attention; we provide a controlled study with learning curves.

Responsible deployment. Following Bragg et al. [5] and Yin et al. [31], we describe the system as isolated sign recognition, document its limitations in a model card [24] and process video locally on the device of the user.

Table 1. Datasets after conversion to the unified format. Durations are medians; “no hand” is the share of frames in which neither hand is detected.

Dataset	Classes	Clips	Signers	Signers tr/val/test	Clips per class	Duration	fps	No hand
ASL (Google ISLR)	250	94,477	21	14/3/4	299–415	0.73 s	30	34 %
LSFB-ISOL	100	55,286	99	50/9/40	216–3293	0.32 s	50	2 %

3 Data

Google ISLR (ASL). The competition data [8] contain 94,477 sequences of 250 ASL signs produced by 21 participants. Each frame provides the 543 MediaPipe Holistic landmarks (468 face, 33 body and 21 per hand); no video is released. Signs were recorded one at a time with a smartphone, in a dictionary-like setting. The landmarks come from the legacy MediaPipe Holistic *solution*, an older pipeline than the MediaPipe Tasks API used in the browser (Section 4.5).

LSFB-ISOL (French Belgian Sign Language). LSFB-ISOL v2 [12] contains 120,739 isolated signs cut from spontaneous conversations of the LSFB corpus [23], covering 4,657 glosses and 99 signers recorded at 50 fps. Landmarks are extracted with the MediaPipe Tasks models (33 body and 478 face points). Since the gloss distribution is long-tailed, we retain the 100 most frequent glosses performed by at least ten signers. We use the *raw* landmarks, without the interpolation and smoothing of the default release, so that missing detections appear as they would during live use. Clips shorter than four frames are discarded (0.6 % of the selected clips, most of them empty). LSFB is related to LSF but is a distinct language; LSFB results are never presented as LSF results.

Unified landmark subset. Both datasets are converted to a common representation of $L = 89$ landmarks per frame: both hands (2×21), seven upper-body points (nose, shoulders, elbows, wrists) and 40 points of the inner and outer lip contours, which carry part of the mouthing information (Section B). Hips are excluded because they usually lie outside the camera field. A sequence is a tensor $X \in \mathbb{R}^{T \times L \times 3}$ of normalised image coordinates (x, y, z) , with NaN for undetected landmarks. The raw ASL data (more than 40 GB of parquet files) are converted once in a cloud notebook into a 2 GB file; LSFB poses are streamed from the dataset server and reduced on the fly.

Aspect ratio. MediaPipe normalises x by the frame width and y by the frame height. Distances are therefore comparable across sources only after multiplying y by the height-to-width ratio of the *displayed* image. LSFB videos are 720×576 PAL frames with a 16:9 display aspect, which inflates vertical distances by $1024/576 \approx 1.78$. We verified this with an anatomical invariant, the height of the nose above the shoulder line divided by the shoulder span: its median is 0.40 in ASL and 0.76 in raw LSFB, and 0.43 after correction (a 4:3 assumption would give 0.57). We therefore scale y by 0.5625 for LSFB, by 1 for ASL, whose coordinates are close to isotropic, and by the actual frame ratio for webcam and video input in the demo. After this correction and the normalisation of Section 4.1, both corpora share the same body frame (Figure 4).

Comparison of the corpora. Figure 1 contrasts the two datasets. ASL classes are balanced and signs are recorded in isolation. LSFB class frequencies span more than an order of magnitude, and clips are shorter (median 0.32 s against 0.73 s) because conversational signs are co-articulated with their neighbours. Missing hands are far more frequent in ASL (Figure 2): one hand is absent from most frames, and neither is detected in 34.1 % of them, against 1.5 % in LSFB. These frames are not idle: in a random sample of 5,000 clips, only 2.4 % of the frames without any hand lie before or after the sign. The others fall within it, when fast motion or the narrow smartphone framing defeats the hand detector. Figure 3 shows one clip of each dataset.

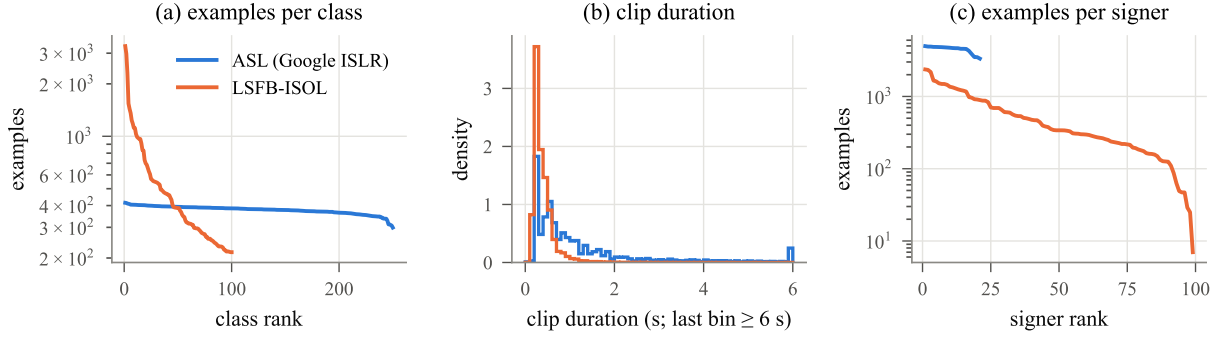


Figure 1. Dataset overview. (a) Examples per class, sorted; (b) distribution of clip durations; (c) examples per signer, sorted.

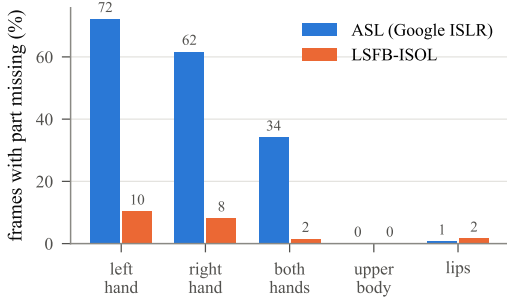


Figure 2. Share of frames in which a body part is not detected.

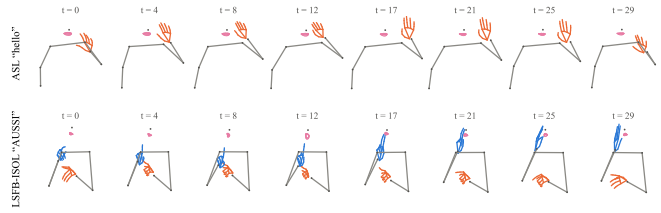


Figure 3. One ASL and one LSFB clip, eight evenly spaced frames (hand slots in orange and blue, lips in pink, upper body in grey).

4 Method

Figure 5 summarises the system. A deterministic preprocessing function maps a raw landmark sequence to a fixed-size feature matrix, which is augmented during training only and classified by a sequence model.

4.1 Preprocessing

Let $X \in \mathbb{R}^{T \times L \times 3}$ be a raw sequence. Preprocessing applies the following steps in order.

Temporal trimming. Leading and trailing frames without any detected hand are removed; a sequence in which no hand is ever detected is kept unchanged.

Canonical dominant hand. One-handed signs are produced with the dominant hand, and the handedness assigned by the landmark estimator depends on whether the image is mirrored. We count the frames in which each hand slot is detected and, if the left slot is more frequent, mirror the sequence: $x \mapsto 1 - x$ for all points, combined with the permutation π that swaps left and right hands, shoulders, elbows, wrists and symmetric lip points. The resulting features are invariant to a horizontal flip of the input whenever one hand dominates, a property verified by a unit test.

Spatial normalisation. Let s_t^l and s_t^r denote the shoulder positions in frame t , and $\mathcal{T}_s$ the set of frames in which both are detected. All points are centred and scaled as

$$c = \frac{1}{|\mathcal{T}_s|} \sum_{t \in \mathcal{T}_s} \frac{s_t^l + s_t^r}{2}, \quad w = \frac{1}{|\mathcal{T}_s|} \sum_{t \in \mathcal{T}_s} \|s_t^l - s_t^r\|_2, \quad \tilde{X}_{t,j} = \frac{X_{t,j} - c}{w}, \quad (1)$$

applied to (x, y) , with z divided by w . Both constants are computed over the whole clip, so that the motion of the hands relative to the body is preserved. If the shoulders are missing or degenerate ($w < 0.02$), the

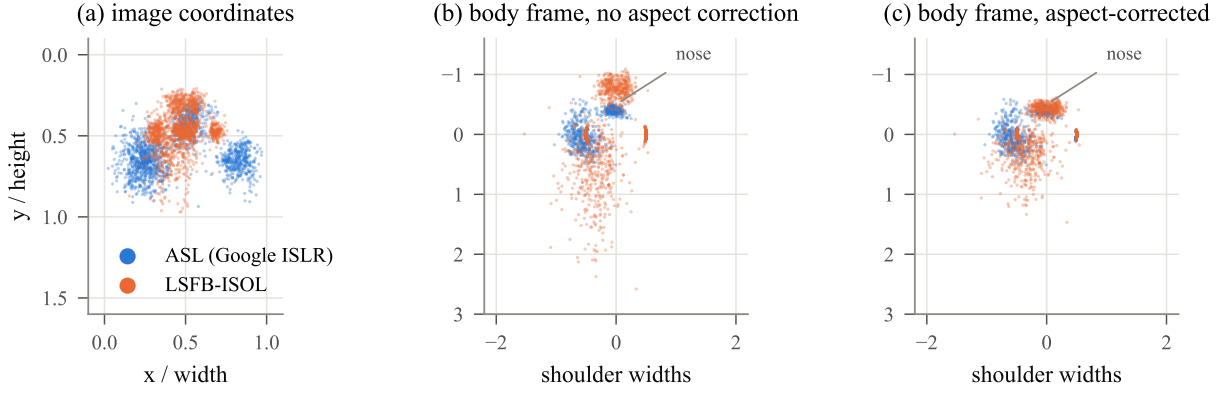


Figure 4. Per-clip mean position of the nose, the shoulders and the dominant wrist for 400 random clips of each corpus. (a) Image coordinates; (b) body frame without aspect correction, where the LSFB nose lies twice as high as the ASL one; (c) body frame with aspect correction, where both corpora align.

in the browser: *MediaPipe Tasks JS + preprocess.js + ONNX Runtime Web*

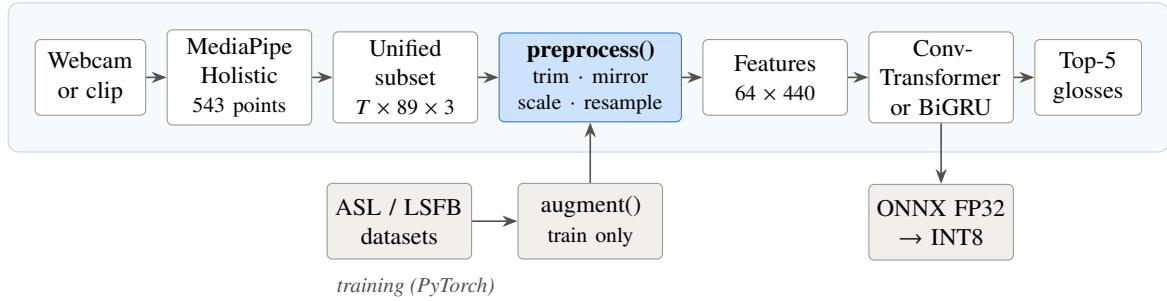


Figure 5. System overview. The same preprocessing function serves training (Python) and the browser runtime (JavaScript port, tested for parity). Only landmarks are processed; video never leaves the device.

centroid of all visible points and four times their mean standard deviation are used instead.

Fixed length. The clip is linearly resampled to $T' = 64$ frames, which also normalises signing speed and frame rate (30 fps for ASL, 50 fps for LSFB). When an interpolation involves a missing neighbour, the value of the nearest frame is used. Zero-padding with a mask is evaluated as an alternative in Section 6.2.

Features. Each frame is described by (i) the normalised (x, y) coordinates of the 89 landmarks; (ii) the dominant hand expressed relative to its wrist and divided by the length of the segment from the wrist to the base of the middle finger, which separates handshape from hand position; and (iii) the first-order temporal differences of (i) and (ii). Depth is discarded by default. This yields $F = 2 \times (89 + 21) \times 2 = 440$ features per frame. Remaining missing values are set to zero, i.e. missing landmarks are encoded at the body centre.

4.2 Data augmentation

Augmentation is applied during training only, in two stages. The first stage acts on the raw sequence: a time warp combining a global speed change ($\sigma \sim \mathcal{U}[0.8, 1.2]$, the clip being resampled to T/σ frames) with a smooth monotonic warp of the time axis over four segments with relative speeds in $[0.7, 1.3]$; random frame dropping ($p = 0.1$); random masking of individual landmarks ($p = 0.05$) and of a whole hand in a frame ($p = 0.05$); and Gaussian noise ($\sigma = 0.002$). The non-uniform component is essential: since every clip is resampled to T' frames, a uniform speed change alone would be almost entirely undone by length normalisation.

The second stage is applied with probability 0.8 *after* shoulder normalisation, in the body frame: a rotation of up to $\pm 15^\circ$, a per-axis scaling in $[0.8, 1.2]$ and a translation of up to 0.1 shoulder widths. Applied in image coordinates, before normalisation, translations and isotropic scalings would be cancelled exactly, as our unit tests confirm. Random draws are seeded by (seed, epoch, sample) for reproducibility.

4.3 Models

Input projection. Both models first project each frame to $d = 192$ dimensions with two linear layers, layer normalisation and GELU activations.

BiGRU. Two bidirectional GRU layers [7] with 192 hidden units per direction, followed by masked mean and max pooling over time and a linear classifier (1.43 M parameters).

Convolution–Transformer. After a learned positional embedding, the network stacks two stages, each made of three convolution blocks and one Transformer block [29]. A convolution block is a pre-norm residual unit: pointwise expansion with a gated linear unit [9], depthwise temporal convolution with kernel size 17, batch normalisation, SiLU and pointwise projection, as in the convolution module of the Conformer [14]. Transformer blocks use four heads and an MLP ratio of two. The sequence is mean-pooled under the mask before the classifier (2.16 M parameters). Attention is implemented with plain matrix products, so that the model exports to ONNX without custom operators. We refer to this model as the Conv-Transformer.

4.4 Training

Models are trained with AdamW [20] (learning rate 10^{-3} , weight decay 0.05, batch size 128), a cosine schedule with three warm-up epochs, label smoothing of 0.1 [26] and gradient clipping at norm 1, for at most 50 epochs. Early stopping monitors the validation macro-F1 with a patience of 10 epochs, and the checkpoint with the best validation macro-F1 is retained. Mixed precision is used on GPU. Section A lists all hyper-parameters.

Transfer. For cross-lingual transfer, all weights except the classifier are initialised from the ASL model, and a new classifier is trained for the target vocabulary. Since the input representation is identical across datasets, no adapter is required. We compare full fine-tuning with a variant that freezes the encoder during the first five epochs. Two additional controls test the robustness of the result: fine-tuning with a five times lower learning rate, and a linear probe in which the encoder stays frozen throughout training.

4.5 Browser runtime

The demo runs the MediaPipe Holistic Landmarker (Tasks API, WebAssembly with GPU delegate) on the webcam stream, builds the 89-landmark frame and passes it to a segmenter that detects the start and end of each sign. The segmenter is a state machine: a sign starts after three consecutive frames with a visible hand and ends when the hands disappear for six frames, remain still (mean hand displacement below 0.012 shoulder widths per frame) for fourteen frames, or after 120 frames. The segment is then preprocessed by the JavaScript port and classified by the ONNX model with ONNX Runtime Web [25] (WebAssembly backend).

Mismatches between training and deployment preprocessing are a common cause of failure in deployed systems. We therefore generate test fixtures from the Python implementation (missing hands, mirrored input, missing shoulders, single-frame and long clips, padding, depth features) and verify that the JavaScript port reproduces the features within 3×10^{-6} , i.e. at single-precision accuracy. We also test the complete browser path: executed in headless Chromium through the code of the demo, the LSFB test clips shipped with the demo receive the same top-1 prediction as with the PyTorch checkpoint (40/40 clips), with probabilities differing by at most 2×10^{-6} .

Table 2. ASL results (250 signs). Validation and test sets contain 3 and 4 signers absent from training. Sequence models: mean $\pm$ standard deviation over three seeds, 50 epochs. The last column is the standard deviation of top-1 accuracy across the four test signers. All values in %.

Model	Params	Validation	Test (4 unseen signers)			
		Top-1	Top-1	Top-5	Macro-F1	Signer s.d.
Chance	–	0.40	0.40	2.0	–	–
Statistics + log. regression	–	47.9	42.1	69.4	41.9	–
Statistics + LightGBM	–	45.2	33.8	61.5	33.4	–
BiGRU	1.43 M	65.8 $\pm$ 0.5	63.0 $\pm$ 0.5	84.0 $\pm$ 0.3	62.3 $\pm$ 0.6	3.9 $\pm$ 0.4
Conv-Transformer	2.16 M	68.2 $\pm$ 0.1	64.8 $\pm$ 0.4	84.0 $\pm$ 0.3	64.1 $\pm$ 0.4	3.3 $\pm$ 0.2

5 Experimental protocol

Splits. All splits are *signer-independent* and frozen in version-controlled files. For ASL, the 21 participants are divided into 14 training, 3 validation and 4 test signers. For LSFB, we keep the official test signers and draw 15 % of the official training signers for validation (Table 1). To measure leakage (RQ1), we also build a random, sample-level ASL split with the same proportions, in which every signer appears in all subsets.

Metrics. We report top-1 and top-5 accuracy, macro-F1 over classes, and the standard deviation of top-1 accuracy across test signers. Model selection and all ablations rely on the validation signers; the test signers are used once, to evaluate the final models.

Statistical tests. Two models evaluated on the same test clips are compared with a paired bootstrap of their accuracy difference [11], using per-clip correctness averaged over the three seeds. The *signer-level* bootstrap draws test signers with replacement and then clips within each drawn signer, so that its 95 % interval accounts for the choice of the test signers; the *clip-level* bootstrap only draws clips within each signer and treats the signers as fixed. We also apply an exact McNemar test [22] to the predictions of the seed ensembles.

Baselines. Besides chance level, we compute per-coordinate temporal statistics (mean, standard deviation, minimum and maximum of the normalised position features over the clip, 4×220 features) and classify them with a multinomial logistic regression or with gradient-boosted trees (LightGBM).

Seeds and compute. Main models are trained with three seeds and reported as mean $\pm$ standard deviation; ablations use a single seed. Deep models are trained on one NVIDIA T4 GPU in a cloud notebook, except for two transfer controls run on the GPU of an Apple M1 laptop once the cloud quota was exhausted. Latency is measured on the laptop CPU with one thread.

6 Results

6.1 ASL: baselines and sequence models

Table 2 reports the main comparison. Temporal statistics are already informative: a logistic regression reaches 42.1 % top-1 on unseen signers, about 105 times chance level. Gradient-boosted trees fit the training signers better but generalise worse (33.8 % on test, 11 points below their validation score, against 6 points for the linear model). With only 14 training signers, flexible models on hand-crafted statistics capture signer-specific cues.

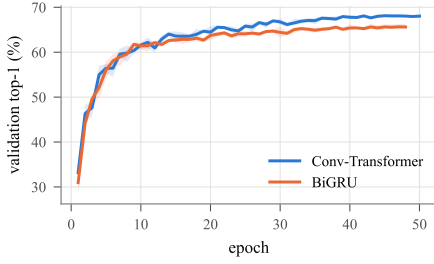


Figure 6. Validation top-1 during training (mean over seeds; band: min–max).

Table 3. Most frequent confusions of the Conv-Transformer on the ASL test signers (seed 0). Rate: share of the clips of the true sign.

True sign	Predicted	Count	Rate
wake	awake	40	51 %
pen	pencil	36	45 %
think	penny	33	43 %
awake	wake	31	42 %
duck	goose	26	32 %
look	see	24	26 %
say	chin	23	32 %
scissors	cut	22	33 %
have	animal	22	31 %
listen	hear	21	22 %
lips	mouth	21	28 %
there	boat	20	27 %

Both sequence models exceed the baselines by more than 20 points. The Conv-Transformer reaches 64.8 ± 0.4 % top-1 on the test signers and the BiGRU 63.0 ± 0.5 %, with the same top-5 accuracy of 84.0 %. Variation across seeds is small (below one point), whereas variation across signers is large: accuracy ranges from 61.9 to 69.5 % over the four test signers, and the Conv-Transformer outperforms the BiGRU on each of them (Figure 7a). The difference of 1.8 points is significant even when the test signers themselves are resampled (signer-level 95 % interval [0.9, 2.9]; clip-level [1.4, 2.2]; McNemar $p < 0.001$), although four signers remain a small sample. Validation accuracy increases until the end of the 50-epoch schedule (Figure 6), yet a three times longer schedule brings no improvement (Section 6.2): the late gains result from the decay of the learning rate.

Error analysis. The most frequent confusions (Table 3) are linguistically plausible. Several pairs are near-synonyms that share an identical or very similar sign in many ASL varieties (WAKE/AWAKE, PEN/PENCIL, LOOK/SEE); others differ by a small change of handshape or location (DUCK/GOOSE, THINK/PENNY). The hardest classes, THERE (3 %), FINGER (12 %), GO (18 %), are pointing and directional signs whose form depends on the location of the referent, which a closed-vocabulary classifier cannot resolve. Part of the residual error is therefore a property of the vocabulary rather than of the model.

Landmark quality is the other main factor (Figure 7b). Accuracy reaches 76.5 % when hands are missing in at most 10 % of the frames, but falls to 52.0 % when no hand is detected in more than half of the frames, which concerns 30 % of the test clips. Clips with a hand visible in every frame are slightly less accurate than the next bin because they are shorter (median of 16 frames instead of 24): among them, clips of at most ten frames, which are probably truncated, are recognised in 59 % of cases, against 77 % for clips of 20 to 40 frames.

6.2 Leakage, ablations and training budget

Leakage. With a random, sample-level split of the same size, in which every signer appears in both training and test data, the Conv-Transformer reaches 81.7 % test top-1 instead of 64.8 %. A random split thus overestimates the accuracy on new users by 16.9 points, more than the effect of any architectural or preprocessing choice studied here.

Ablations. Figure 8 changes one factor at a time with respect to the full model (68.3 % validation top-1; seed-to-seed standard deviation of 0.1 point). Early stopping proved to be a confounding factor. Two variants initially stopped on a validation plateau after 21 and 31 epochs, while the learning rate was still high, and appeared to lose 4.4 and 2.7 points. Re-run over the full 50-epoch schedule, their effects reduce

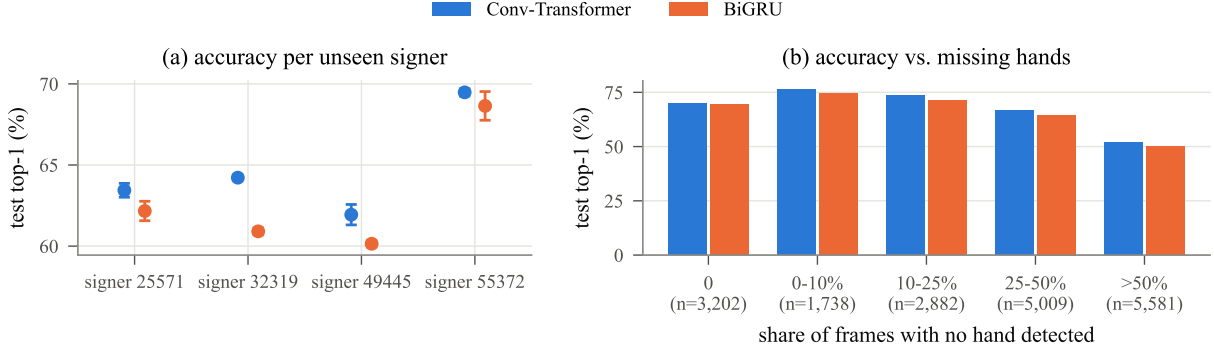


Figure 7. (a) Test top-1 for each unseen signer (mean and standard deviation over seeds). (b) Test top-1 as a function of the share of frames in which neither hand is detected.

to -0.8 (no hand-local features) and -1.1 (depth added) points. The other variants marked with † (4 in total) stopped between epochs 41 and 46; their effects may be slightly overestimated, and our compute budget did not allow re-running them.

Under this caveat, the canonical dominant hand has the largest effect (-3.8 points without it): otherwise, the same sign occupies either hand slot depending on the handedness of the signer and on mirroring. Velocities (-1.7), augmentation (-1.3), depth (-1.1, consistent with the noise of monocular depth estimates) and the hand-local shape features (-0.8) have moderate effects.

Three findings depart from common practice. First, the lips and the upper body bring no measurable gain on this vocabulary (+0.5 without lips, +0.0 with hands only): the 250 signs are mainly distinguished by the hands, and mouthing is presumably rare in dictionary-like recordings. Second, shorter sequences perform as well ($T' = 32$: +0.2) and longer ones no better ($T' = 96$: -0.8), the median clip containing only 22 frames. Third, zero-padding with a mask outperforms resampling (+1.2): resampling stretches short clips and removes the duration cue, whereas padding preserves the original temporal dynamics. The resampled configuration, fixed before the ablations, is kept in the rest of the study, but padding is the better choice for future work.

Training budget and ensembles. Training the Conv-Transformer for 150 instead of 50 epochs does not help (67.5 % against 68.3 % on validation): the late rise of the curves in Figure 6 reflects the annealing of the learning rate, not under-training. Averaging the predictions of the three seeds, by contrast, improves test top-1 to 67.4 % on ASL (64.8 % for a single model) and to 70.9 % on LSFB (68.3 %), at three times the inference cost.

6.3 Cross-lingual transfer to LSFB

Table 4 reports the results on LSFB-ISOL with all training data. Trained from scratch, the Conv-Transformer reaches 68.3 % top-1 and 63.9 % macro-F1 on the 40 official test signers, well above the statistical baselines (49.0 % and 40.9 %). Macro-F1 is markedly lower than top-1 on this corpus because of its long tail: the ten most frequent glosses account for 32 % of the test clips.

Validation and test accuracy. On LSFB, validation accuracy exceeds test accuracy by far more than on ASL (77.2 against 68.3 %). The validation set is small and unbalanced across signers: three of its 9 signers contribute 75 % of its clips. Averaged per signer, accuracy is 72.7 ± 2.5 % on validation (seed 0) and 67.9 ± 2.0 % on test (mean over seeds; mean $\pm$ standard error over signers), and re-weighting the test clips to the class distribution of the validation set raises test accuracy to 70.2 %. LSFB clips come from dialogues: test signers whose conversation partner belongs to the training signers (17 of 40) are recognised with 71.8 % accuracy, the others with 64.7 %, which suggests that shared conversations, and therefore shared topics and signing styles, also favour held-out signers. Validation scores are therefore used only to compare configurations, never as an estimate of performance on new signers.

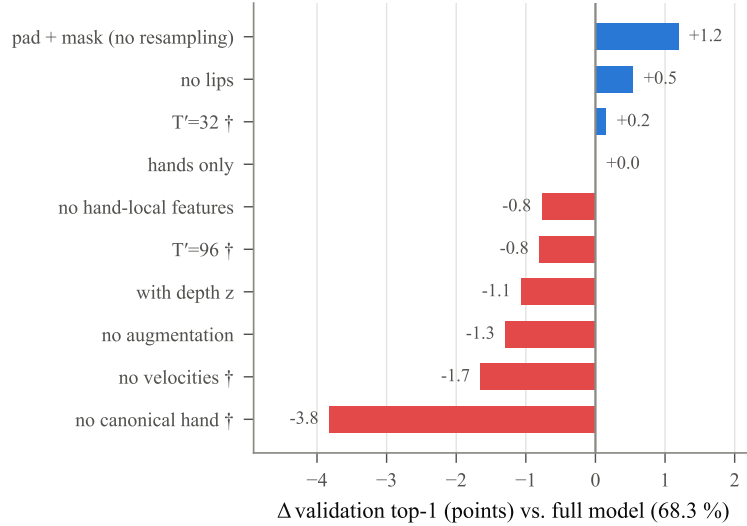


Figure 8. One-factor ablations of the Conv-Transformer on the ASL validation signers (seed 0). †: early stopping ended the run before the end of the 50-epoch schedule (epochs 41–46), so the effect may be overestimated.

ASL pre-training. Initialising the encoder from the ASL model does not improve recognition. With all training data, fine-tuning the ASL model yields 66.2 % (74.9 % on validation, against 77.0 % from scratch), and freezing the encoder during the first five epochs only narrows the gap (66.6 %). Pre-training is not merely useless but slightly harmful: relative to training from scratch, test accuracy changes by -2.1 points with fine-tuning (signer-level 95 % interval [-2.8, -1.4], $p < 0.001$) and by -1.7 points with the frozen variant ([-2.3, -1.2]). The learning curves of Figure 9 show that this result does not depend on the size of the training set. Both initialisations are indistinguishable from 5 to 25 examples per sign (the largest difference, 1.4 points at 5 examples, lies within one seed standard deviation), and the model trained from scratch leads from 50 examples per sign onwards, by 2.0 points with all data.

Controls. Two controls exclude an inadequate fine-tuning procedure as the explanation. First, a linear probe trained on top of the frozen ASL encoder with all LSFB training data reaches 38.3 % validation top-1, far below the model trained from scratch (77.0 %) and even below the logistic regression on temporal statistics (57.0 %). The ASL representation therefore retains some information about LSFB signs, but less than simple features of the target data. Second, with 25 examples per sign, fine-tuning with a five times lower learning rate gives 46.3 % validation top-1, below both standard fine-tuning (50.6 %) and training from scratch (51.5 %) with the same seed. Both controls use a single seed, and every transfer run starts from the ASL model of seed 0, so the variability of the pre-trained model itself is not sampled. During the linear probe, the batch-normalisation statistics of the frozen encoder were still updated on LSFB, a mild form of adaptation; the probe is therefore, if anything, favoured, which does not change its conclusion.

6.4 Efficiency and deployment

Latency. Table 5 summarises the inference cost. Preprocessing takes about 0.3 ms per clip in Python, and each sign is classified once, when the segmenter detects its end. The Conv-Transformer then requires 7.8 ms natively and 16 ms in the browser, and the BiGRU 8 ms in the browser. Landmark extraction by MediaPipe, which runs on every frame, dominates the cost of the live demo.

Quantisation. Naive dynamic INT8 quantisation reduces the validation top-1 of the Conv-Transformer from 68 % to 34 %. The cause is a small number of very large inputs: when a detected hand is tiny, the hand-local features (Section 4.1) are divided by a near-zero length and reach values close to 200, while 99.9 % of them remain below 6.2. A per-tensor dynamic scale on the first layer then destroys the resolution of all other inputs. Keeping this single layer in FP32 fully restores accuracy. The quantised model is 3.3×

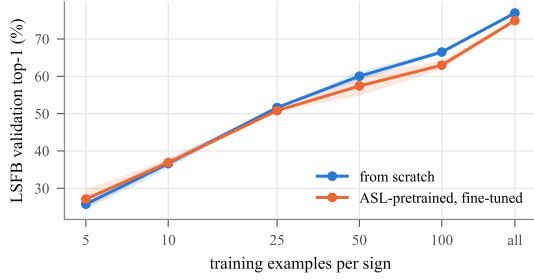


Figure 9. LSFB validation top-1 as a function of the number of training examples per sign (mean $\pm$ s.d. over three seeds). “All” corresponds to a median of 185 examples per sign.

Table 4. LSFB-ISOL, 100 glosses, all training data. Mean $\pm$ s.d. over three seeds, values in %.

Model	Validation	Test (40 unseen signers)		
	Top-1	Top-1	Top-5	Macro-F1
Chance	1.0	1.0	5.0	–
Statistics + log. regression	57.0	49.0	75.3	40.9
Statistics + LightGBM	54.8	47.1	72.6	37.3
Conv-Transformer, from scratch	77.0 $\pm$ 0.3	68.3 $\pm$ 0.2	87.7 $\pm$ 0.1	63.9 $\pm$ 0.2
ASL-pretrained, fine-tuned	74.9 $\pm$ 0.4	66.2 $\pm$ 0.2	87.1 $\pm$ 0.4	61.7 $\pm$ 0.3
ASL-pretrained, encoder frozen 5 epochs	75.6 $\pm$ 0.6	66.6 $\pm$ 0.2	86.5 $\pm$ 0.5	62.0 $\pm$ 0.3

Table 5. Model size and single-sign latency on an Apple M1 CPU (one thread) after export to ONNX. Native: ONNX Runtime; browser: ONNX Runtime Web (WebAssembly) in Chromium. INT8: dynamic quantisation of the weight matrices, with the input projection kept in FP32. Validation top-1 on 2,000 ASL validation clips.

Model	Precision	Size	Native	Browser (WebAssembly)		Val. top-1
			p50 (ms)	p50 (ms)	p95 (ms)	
Conv-Transformer	FP32	8.3 MB	7.8	16.3	22.0	67.8
Conv-Transformer	INT8	2.5 MB	16.5	33.4	44.8	68.3
BiGRU	FP32	5.5 MB	5.7	7.9	11.0	–
BiGRU	INT8	4.8 MB	5.5	8.1	10.2	–

smaller but $2.1\times$ *slower* on CPU, natively and in WebAssembly: at this scale, quantising and dequantising activations around many small matrix products costs more than integer arithmetic saves. Since the demo already downloads about 25 MB for MediaPipe, it ships the FP32 model.

7 Discussion

Sources of error. All results are obtained on people absent from training, a setting that Section 6.2 shows to be considerably harder than a random split suggests. The remaining errors have three main sources, which call for different remedies: signs that are genuinely ambiguous within a closed vocabulary (near-synonyms, pointing and directional signs); landmark detection failures, which dominate the hardest clips; and variation between signers, which exceeds the difference between the two architectures. More signers and better hand tracking are therefore likely to help more than a larger classifier.

Absence of transfer from ASL. We expected an ASL encoder to help, at least with few LSFB examples, since both languages build signs from handshapes, movements and locations. The two corpora, however, differ in more than the language. ASL clips are citation forms recorded one at a time on a smartphone; LSFB clips are extracted from spontaneous conversations filmed in a studio, are less than half as long (median 0.32 s against 0.73 s) and are co-articulated with neighbouring signs. Hands are missing in 34.1 % of the ASL frames but in 1.5 % of the LSFB frames, and the landmarks come from two generations of MediaPipe. The pre-trained encoder is therefore adapted to a different input distribution, and even a few hundred labelled clips of the target corpus provide more relevant information than this prior. This explanation is consistent with our measurements but not demonstrated: separating the effect of the language from that of the recording conditions would require a second ASL corpus of conversational signs, or an LSFB corpus of citation forms. Self-supervised pre-training on unlabelled, in-domain continuous signing,

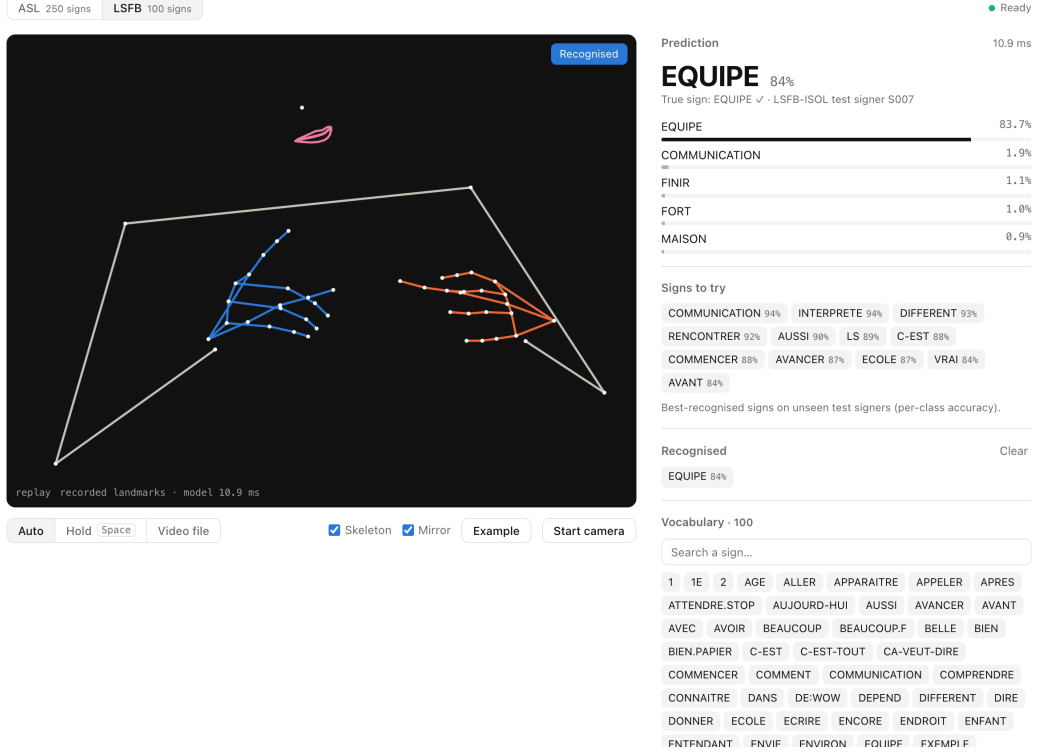


Figure 10. Browser demo replaying an LSFB-ISOL clip of an unseen test signer: skeleton, top-5 predictions with the true sign, best-recognised signs, history and vocabulary. The demo ships one test clip for each of 40 glosses, of which the model recognises 70.0 %. With a webcam, signs are segmented automatically from hand presence and motion.

for instance LSFB-CONT [12] or MEDI-API-SKEL [6] for French Sign Language, appears to be a more promising direction than supervised cross-lingual pre-training.

Methodological pitfalls. Several of our observations are inexpensive to check yet easily overlooked. (i) Image-space translation and scaling augmentations are cancelled by body-centred normalisation and must be applied after it. (ii) Normalised landmark coordinates are anisotropic whenever pixels or frames are not square, which distorts the body frame across datasets. (iii) Early stopping on a plateau while the learning rate is still high can make an ablation appear much more harmful than it is. (iv) Rare feature outliers, harmless in floating point, can break post-training quantisation, and quantisation does not necessarily make a small model faster on CPU. (v) A preprocessing function implemented in two languages must be tested for parity, which costs little once fixtures are generated from the reference implementation.

Limitations. (i) The ASL test set contains only four signers, so per-signer estimates are uncertain, and all LSFB clips come from a single corpus recorded in one studio setting. (ii) Both vocabularies are small and fixed: an out-of-vocabulary sign is always mapped to the closest known sign, and the demo has no rejection class. (iii) Our ASL accuracy remains well below that of the best competition solutions [8], which relied on stronger regularisation and augmentation, much longer schedules and ensembles of several models, and were trained on all 21 participants, whereas our models see 14 of them and are tested on a different set. We did not measure how much each of these factors contributes. A longer schedule alone did not help our model, and the ablations used a single seed, 4 of them ending before the full schedule; differences of about one point between ablations should therefore not be over-interpreted. (iv) Because every clip is stretched to 64 frames, velocities are expressed per resampled frame: their scale depends on the duration of the clip and on the frame rate of the source (30 or 50 fps), which may partly explain why padding performed better in the ablations. Resampling to a common frame rate and padding would

remove this dependence. (v) The lip contour is the only non-manual information used; eyebrows, gaze and head movements, which carry grammatical information, are ignored. (vi) The demo segments signs from hand presence and motion, which suits signs produced one at a time with a pause but not natural, continuous signing. (vii) No Deaf signer took part in the evaluation of the demo; its usefulness for the community remains to be established with its members.

Ethical considerations. Both datasets were released for research by their creators, under the Kaggle competition rules and the CC BY 4.0 licence (LSFB-ISOL). ASL data are never redistributed; the demo ships a few LSFB clips with attribution. The demo processes video locally and stores nothing. Face and body landmarks can nonetheless identify a person, so any extension that records them should obtain explicit consent. We describe the system as *isolated sign recognition*: presenting such a classifier as a translator would mislead users about its capabilities and could cause harm where communication matters. A model card summarising intended uses, out-of-scope uses and known biases accompanies the code [24].

8 Conclusion

We built and evaluated an isolated sign recognition system based on MediaPipe landmarks, under a signer-independent protocol and with a fully client-side deployment. The three research questions receive the following answers.

RQ1 Generalisation. A 2.2 M-parameter convolution–Transformer recognises 250 ASL signs produced by unseen signers with $64.8 \pm 0.4\%$ top-1 accuracy, and 100 LSFB glosses with 68.3 %. A random split inflates the ASL accuracy by 16.9 points, and an early-stopping artefact had overstated the effect of one ablation fivefold: evaluation choices influence conclusions more than most design choices.

RQ2 Transfer. Pre-training on ASL does not improve LSFB recognition at any training-set size, even with five examples per sign, and with all training data it lowers test accuracy by a small but significant margin. Differences in recording and elicitation conditions likely outweigh what the two languages share at the level of pose features.

RQ3 Deployment. The complete pipeline runs in a web browser, classifies a sign in 16 ms on one CPU thread and reproduces the predictions of the PyTorch model on every tested clip. INT8 quantisation reduces model size but not latency on CPU.

Future work includes self-supervised pre-training on in-domain continuous signing, a rejection class for out-of-vocabulary signs, richer non-manual features, and an evaluation of the demo with Deaf signers.

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A Hyper-parameters

B Landmark subset

Table 6. Hyper-parameters shared by all experiments unless stated otherwise. All values are read from the configuration files released with the code.

Group	Parameter	Value
Preprocessing	frames T' / length mode	64 / linear resampling
	coordinates	(x, y) ; depth dropped
	landmark parts	both hands, upper body (7 points), lips (40 points)
	features per frame	positions, dominant-hand local shape, velocities ($F = 440$)
Augmentation	spatial (body frame), probability	0.8
	rotation / per-axis scale / shift	$\pm 15^\circ$ / $[0.8, 1.2]$ / ± 0.1 shoulder widths
	global speed / local warp	$[0.8, 1.2]$ / $\pm 30\%$ over 4 segments
	frame / landmark / hand dropping	0.1 / 0.05 / 0.05
	Gaussian noise (image units)	0.002
BiGRU	input projection / hidden / layers	192 / 192 per direction / 2
	pooling / dropout	masked mean max / 0.3
Conv-Transformer	width / stages	192 / 2
	per stage	3 conv. blocks + 1 Transformer block
	kernel / expansion / heads / MLP ratio	17 / 2 / 4 / 2
	dropout	0.2
Optimisation	optimiser	AdamW, lr 10^{-3} , weight decay 0.05
	schedule	3 warm-up epochs, cosine decay
	batch / max. epochs / patience	128 / 50 / 10 (validation macro-F1)
	label smoothing / gradient clipping	0.1 / 1.0
	precision	mixed (fp16) on GPU

Table 7. Unified landmark subset, in the order used by the model (MediaPipe indices).

Part	Points	MediaPipe indices
Left hand	21	hand model 0–20
Right hand	21	hand model 0–20
Upper body	7	pose 0 (nose), 11–12 (shoulders), 13–14 (elbows), 15–16 (wrists)
Lips	40	face mesh: outer contour (20) and inner contour (20)
Total	89	